

GOAL: Estimate correlations to assess biomarkers

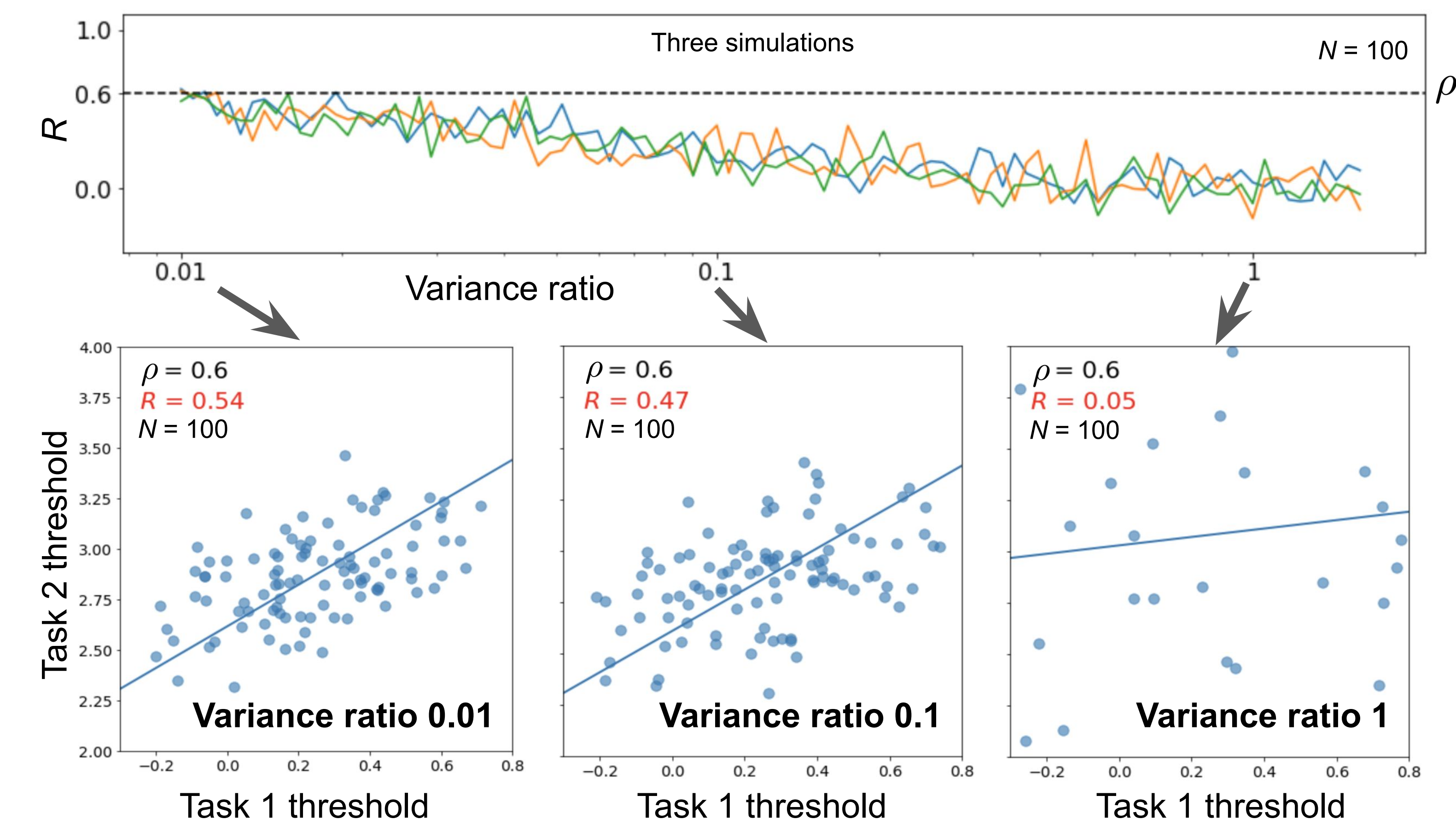
Visual crowding distance: spacing needed to recognize an object in clutter

- Correlates with V4 size [1], informational masking (“auditory crowding”) [2], and reading speed [3].
- We want to assess crowding’s utility as a biomarker. Toward this goal, **this project aims to improve correlation estimates.**

ISSUE 1: Measurement noise

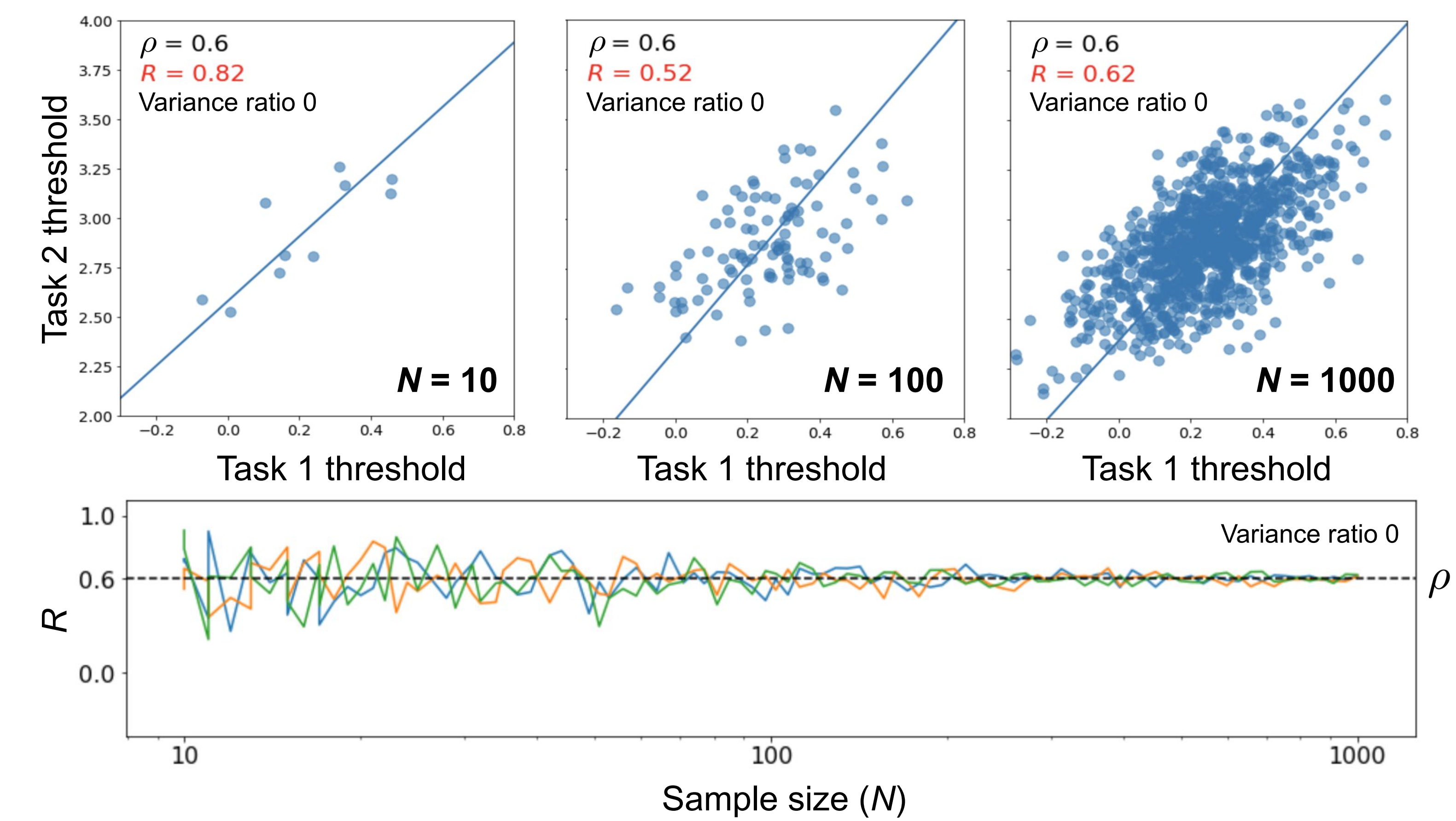
Increasing the **variance ratio** attenuates the estimate R of the true correlation ρ (here, 0.6). **Variance ratio:** $\sigma^2_{\text{thresh}} / \sigma^2_{\text{obs}}$

σ^2_{thresh} = variance across thresholds *within* each observer (i.e., noise)
 σ^2_{obs} = variance of the true thresholds across observers

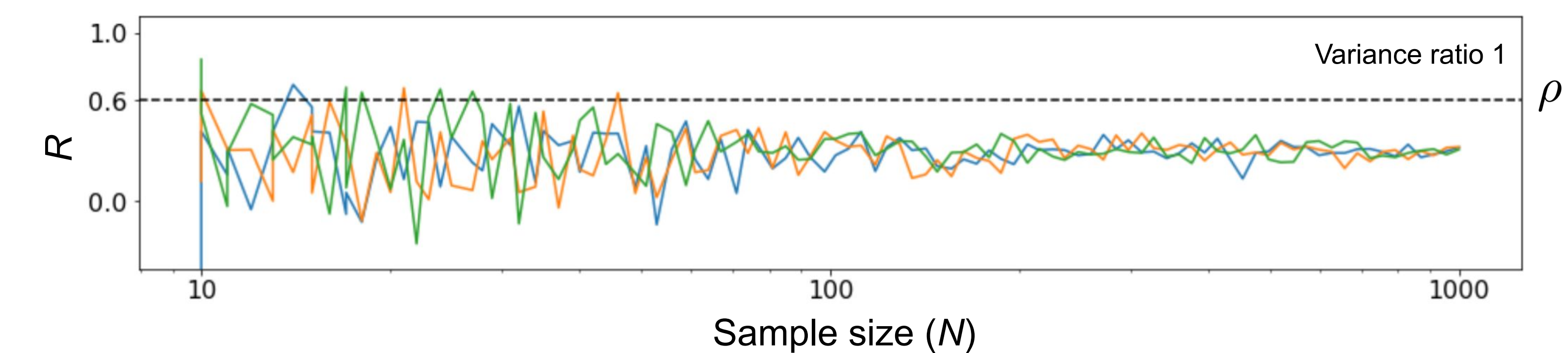


ISSUE 2: Small sample size

Without measurement noise, as N increases, R converges to ρ .



With measurement noise, R converges to a value less than ρ .

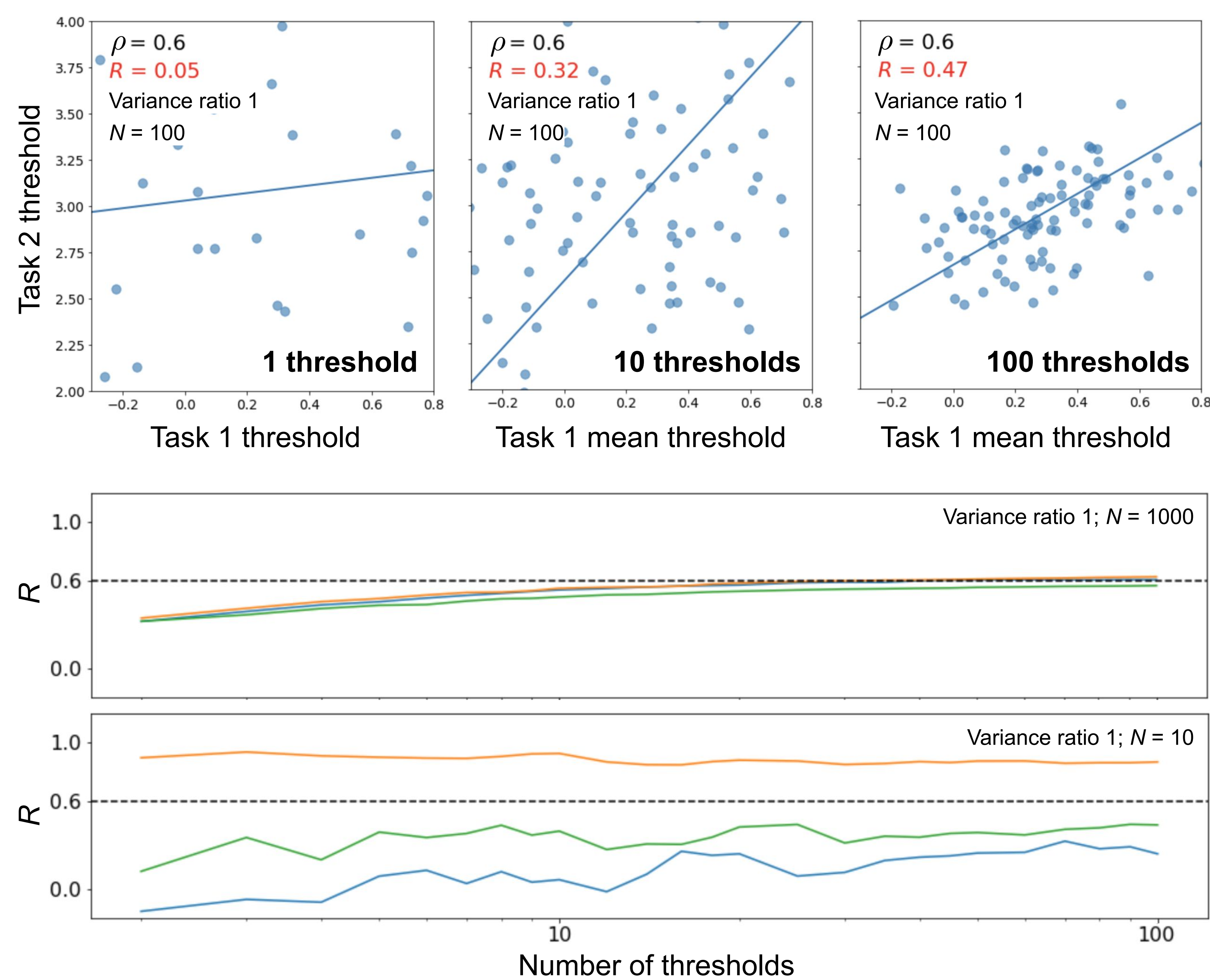


REMEDIES FOR NOISE:

1. Multiple thresholds per observer

Measuring more thresholds lessens the attenuation.

Taking the mean of more thresholds for each observer reduces the effective variance ratio (i.e., variance ratio of the mean of N thresholds).



2. Unbiased estimator

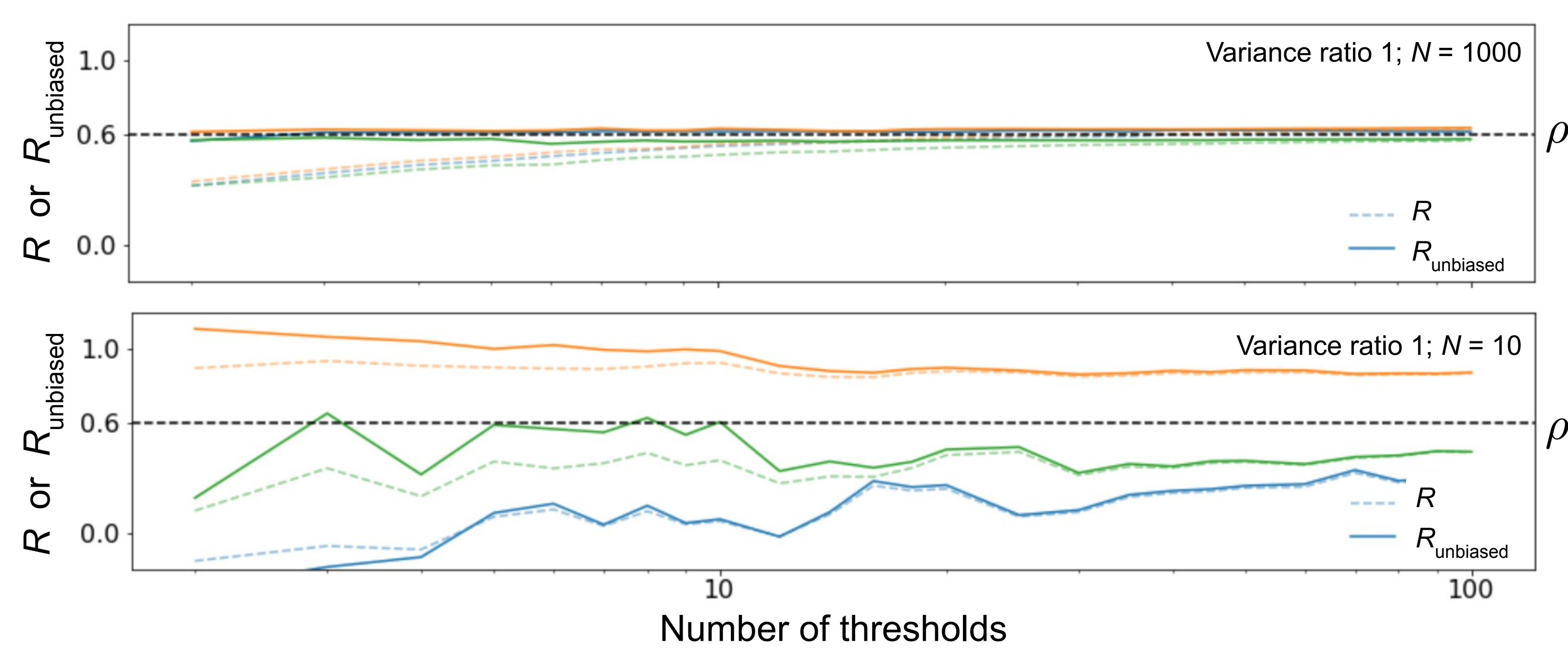
Attenuation is a bias. Spearman (1904) proved that the following correlation estimator is unbiased:

$$R_{\text{unbiased}} = \frac{R}{R_{\text{ceiling}}} = \frac{\text{Corr}(X_{\text{avg}}, Y_{\text{avg}})}{\sqrt{\text{Corr}(X_{\text{test}}, X_{\text{retest}}) \text{Corr}(Y_{\text{test}}, Y_{\text{retest}})}} \quad [4]$$

X and Y : noisy thresholds, each measured twice (test and retest)
 X_{avg} and Y_{avg} : average of test and retest
 Corr : Pearson correlation coefficient
 R_{ceiling} : joint test-retest reliability of X and Y

R_{unbiased} has less bias but more variance than R (since R_{ceiling} is noisy).

- When N is 1000 (upper panel), R_{unbiased} (solid lines) is better than R (dashed lines).
- When N is 10 (lower panel), R_{unbiased} has high variance so R is better.



When is the unbiased estimator better than the raw estimate?

The expected errors of R and R_{unbiased} depend on:

- Variance ratio
 - Ground truth correlation
 - Number of observers
 - Number of thresholds
- Beyond experimenter's control**
In experimenter's control

We want to use the estimator (R or R_{unbiased}) with least root-mean-squared error (RMSE). RMSE accounts for both bias and variance and is the square root of MSE.

$$\text{MSE} = \mathbb{E}[(R - \rho)^2] = \underbrace{(\mathbb{E}[R] - \rho)^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}[(R - \mathbb{E}[R])^2]}_{\text{Variance}}$$

Estimated correlation Ground truth correlation

To choose the least-RMSE estimator:

- Collect pilot data (test and retest for both tasks)
- Compute R_{ceiling} and R (to estimate ρ)
- Compute and compare RMSEs of R and R_{unbiased} by simulation or analytic approximation (below)

$$\text{MSE}_{\text{raw}} \approx \text{Var}(R) + [\rho(1 - R_{\text{ceiling}})]^2$$

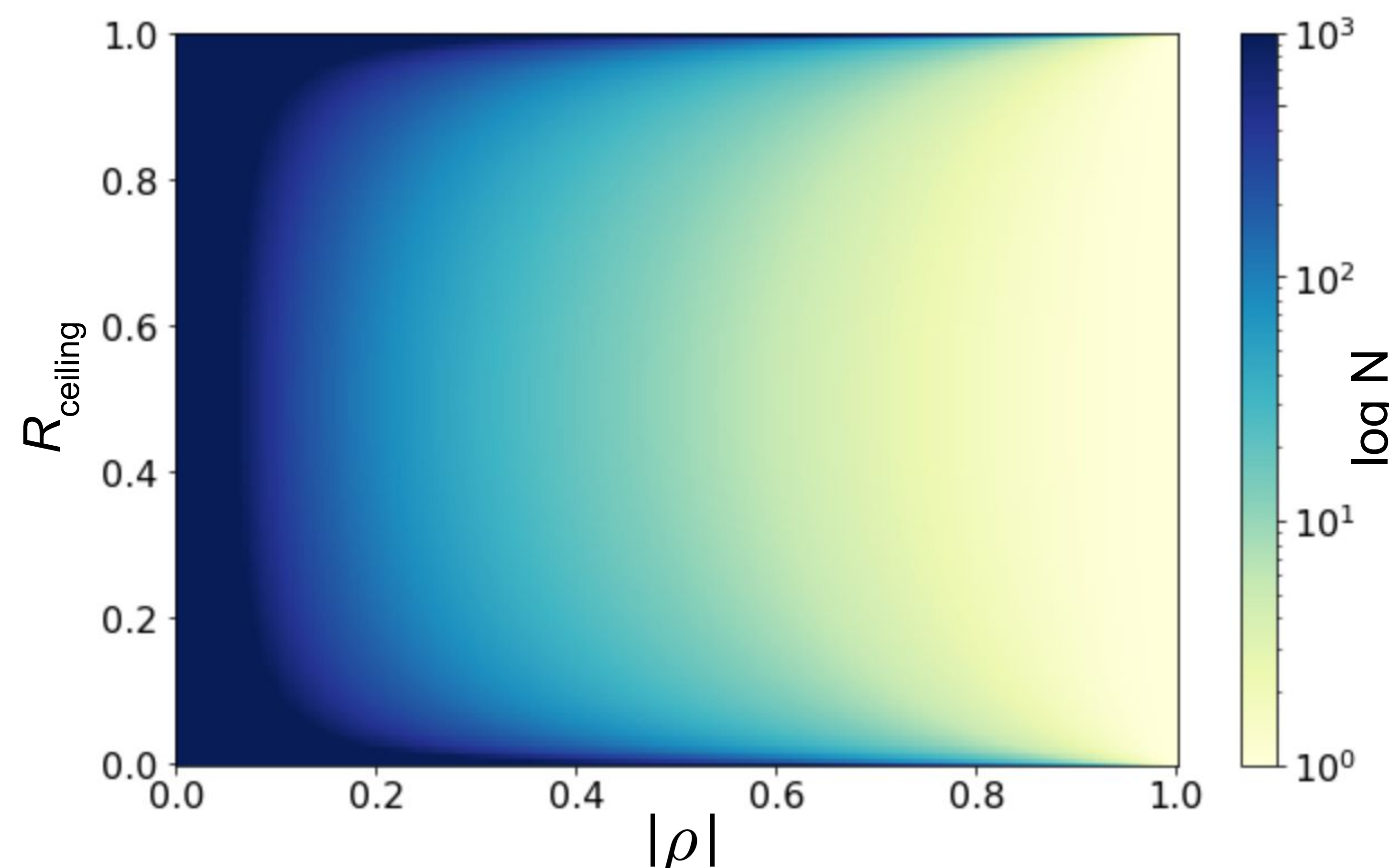
$$\text{MSE}_{\text{unbiased}} \approx \frac{\text{Var}(R)}{R_{\text{ceiling}}}, \text{ where } \text{Var}(R) \approx \frac{(1 - \rho^2)^2}{N - 1} \quad [5]$$

We want to know when $\text{MSE}_{\text{unbiased}} < \text{MSE}_{\text{raw}}$

Plugging in and solving for N shows that we choose R_{unbiased} over R when

$$N > 1 + \frac{(1 - \rho^2)^2 / \rho^2}{R_{\text{ceiling}} (1 - R_{\text{ceiling}})}, \text{ which we plot below.}$$

Sample size required for R_{unbiased} to have lower RMSE than R :



In the light areas, use R_{unbiased} even if N is small.

In the dark areas, use R unless N is very big.

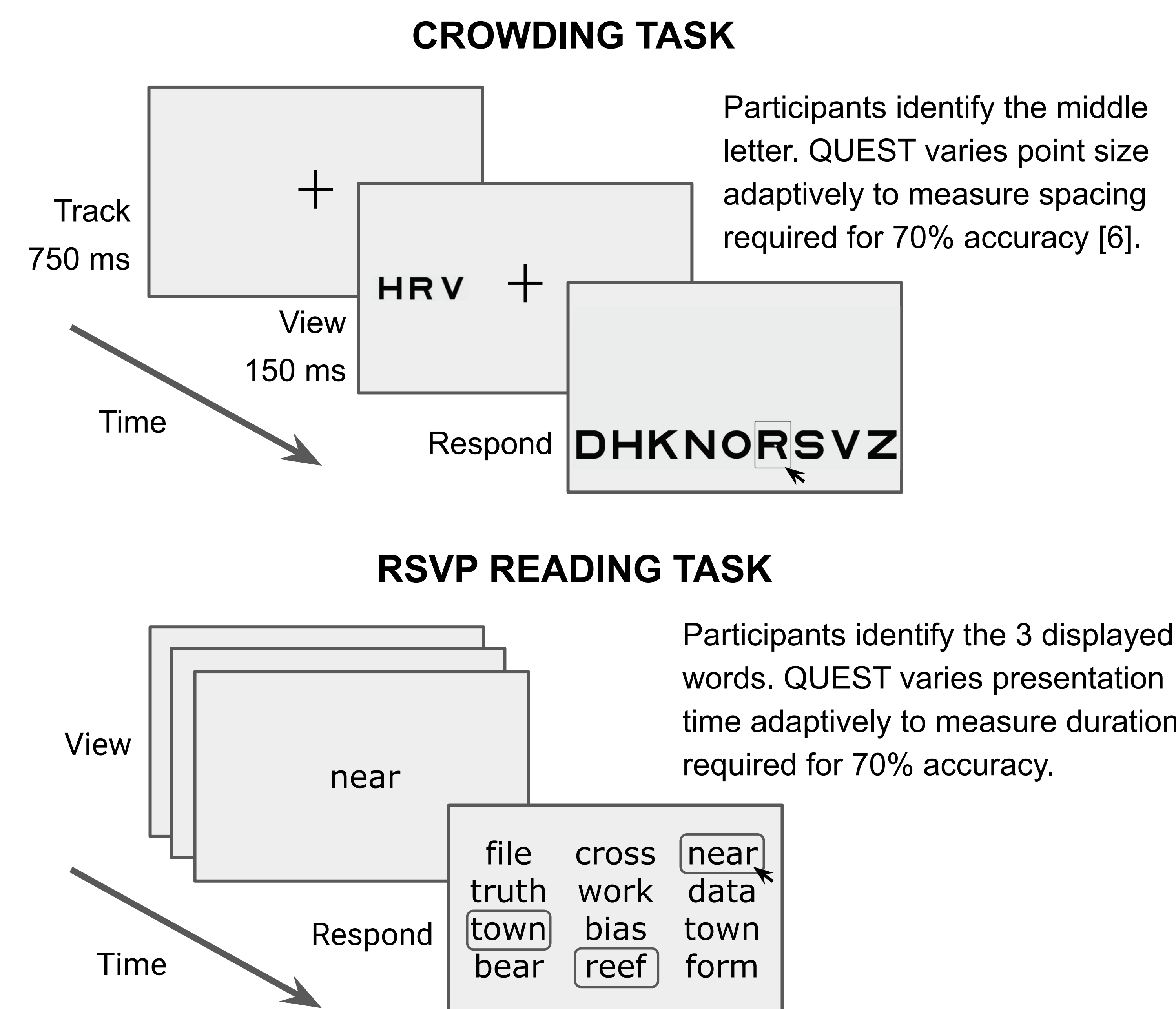
DATA COLLECTION

Pilot data show that crowding and RSVP reading speed have:

- Variance ratios 0.6 and 0.25, respectively
- R_{ceiling} (joint reliability) of about 0.7
- Estimated ρ of 0.65

We plan to collect the following thresholds on 200 online observers (more than enough to justify using the unbiased estimator) using EasyEyes.app (test and retest, in two 1-hour sessions):

- Visual crowding (± 8 deg)
- Letter acuity (± 8 deg)
- RSVP reading speed (foveal)
- Vernier acuity (± 8 deg)
- Ordinary reading speed



EXPECTED OUTCOMES

Pilot data and this procedure show that our full study will benefit from the unbiased estimator.

We anticipate that the correlation between crowding and RSVP will improve from a raw value of **−0.45** to an unbiased estimate of **−0.65**.

This procedure helps quantify the correlation between crowding and reading speed, and we hope it will aid other studies that use correlation.

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CITATION

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